1 Once-counted distinct-port three-block cancellation

For the unchanged frozen propagator, put $P_{xy} = |U_{xy}|^2$ and $J_{xy} = N^{-1}$. The position-chain theorem supplies, for all sufficiently large odd N, $\max P_{xy} \le C/N$ and $\|P^L - J\| \le 2^{-L}$ for every integer $L \ge 0$. Let T_m be the set of rooted ordered coordinate-word pairs obtained by cutting the first cyclic word into three nonempty blocks A, B, C with pairwise distinct initial vertex values, then taking a cyclic rotation of ACB for the second word. This is a set: each ordered pair is counted once, even if several cuts produce it. Write $\mathcal{A}(x) = \prod_{j \bmod m} U_{x_j x_{j+1}}$.

Theorem (BGS-TRACE-015). For every smooth w compactly supported in $(0, \infty)$,

$$\frac{1}{N^2} \sum_m w(m/N) \sum_{(x,y) \in T_m} \mathcal{A}(x)\overline{\mathcal{A}(y)} \longrightarrow 0.$$

For $K = \lceil \log N \rceil$ and support in $[a, b] \subset (0, \infty)$, the error is bounded by

$$O_w(K^5/N) + O_w\Big(N^8 (C/N)^{K/2}\Big) + O_w\Big(N^5 2^{-(aN-3)/3}\Big).$$

The startup threshold inherited from the position-chain theorem is not made effective.

Proof mechanism. In the marked three-link contraction, replacing any link by J gives zero by column orthogonality and distinct ports. Using the longest link therefore suppresses the full marked sum exponentially, including zero internal link lengths. Passing to a once-counted family is a separate combinatorial step. Words with repeated length-K substrings impose at least $K/2$ equality constraints and have superpolynomially small exceptional absolute weight. For the remaining words, two blocks of length at least $4K$ force a unique cut triple, using uniquely located common substring anchors. Ambiguity is consequently confined to two short blocks. Competing cuts lie in an $O(K)$ neighborhood, giving $O(K^3)$ alternatives and $O(K^2)$ original length choices. A rank-one relative positional permutation would be a transposition; conjugating predecessor cycles then forces equal-valued positions to be two cut heads, contradicting distinct ports. Thus every competing representation imposes at least two equalities. Exposing the three starts first pays each equality through a transition bounded by C/N. After the two trace roots and band normalization, the exceptional mass is $O(K^5/N)$. Every nonexceptional pair has exactly three marked representations, corresponding to cyclic block labels. The canonical proof gives all equality, cyclic-root and overlap counts explicitly.

Boundary. This is a deterministic, once-counted, fixed-complexity family estimate. The combinatorial theorem uses only unitarity, flat entries and the

stated position-chain mixing bound; actual-model input enters through BGS-FM-021. It is compatible with spectrally nonuniversal families satisfying those matrix estimates. It does not cover repeated ports, arbitrary reconnections, growing encounter complexity or higher joint traces. The complete signed remainder, including restoration of the super-Heisenberg plateau, remains open; no BGS or local-statistics theorem follows from this family alone.